

A RECURRENT NEURAL NETWORK WITH GENETIC ALGORITHM FOR FLIGHT DELAY CLASSIFICATION

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Abstract

The aviation industry has experienced constant growth in air traffic since the deregulation of the U.S. airline industry in 1978. As a result, flight delays have become a major concern for airlines and passengers, leading to significant research on factors affecting flight delays such as departure, arrival, and total delays. Flight delays result in increased consumption of limited resources such as fuel, labour, and capital, and are expected to increase in the coming decades. To address the flight delay problem, this research proposes an approach that combines the feature of LSTM-RNN and a Genetic Algorithm. In addition, several machine learning algorithms are applied on flight data to validate the results of proposed model. To measure the performance of the model, accuracy, precision, recall, and F1-score are calculated.

I. Introduction

Flight Schedule is essentially planning, mapping out and understanding all of the criteria, materials and routes of a flight that you are undertaking. It is vital that all flights are deemed to have a well-structured and organized flight plan, which will not only maximize the safety of those on board but also reduce costs where necessary. As circumstances are never the same for any flight, the calculations must always be tailored to the flight in question (Alishashi et al., 2021).

Supporting this observation, the 2019 report from the International Civil Aviation Organization (ICAO) indicates a clear and continuous upward trend in the aviation sector. The report highlights substantial growth in air transport activities, with a notable increase in the number of commercial passengers. Specifically, between 2018 and 2019, the aviation industry witnessed an impressive rise, with an additional 155 million commercial passengers taking to the skies. This remarkable growth resulted in an average year-on-year increase of approximately 6.07%. These figures not only underscore the robust demand for air travel but

also reflect broader economic and social trends that have fostered the expansion of the aviation industry.

ML and DL algorithms that have been applied in similar projects or studies with the same aim include traditional Recurrent Neural Network (RNN), Feedforward Neural Networks (FNN), Artificial Neural Networks (ANN), Deep Neural Networks (DNN), Natural Language Processor (NLP), Long Short-Term with Attention Mechanism (LSTM-AM), Convolutional Neural Network (CNN) with LSTM.

The primary objective of this paper is to develop an improved supervised machine learning model capable of accurately classifying flights as either Delayed or On-Time. This paper will make use of the RNN algorithm of the LSTM with a Genetic Algorithm (GA) variant which is best suited for time series data from flight schedules, weather conditions and other relevant factors. To measure the performance of the model, accuracy, precision, recall, and F1-score are calculated.

The organization of the paper is structured in the following way: Section Two is going to talk about the reviewed paper on the existing and related works. Section Three examines the flight delay problem, analysing the flight dataset and presenting various plots to showcase the key insights. Section Four will outline the steps, procedure, process and rules used in carrying out the project. Section Five will give the implementation results of the work and lastly, Section Six, it talks about the discussion of the results, conclusion and recommendation of a possible future work/study.

II. Literature Review

Previous research has used and improved on various algorithms in a bid to efficiently predict the possibility of flight delay outcomes. Bisandu et al. (2022) clearly explained that precise and reliable prediction of flight delay propagation at a national network scale is critically important, particularly for flights that experience recurring delays. Recent advancements in the architectural design of RNN units have led to significant breakthroughs, with LSTM models showing great promise in overcoming these challenges. LSTM models excel at learning and retaining long-term dependencies in sequential data, making them well-suited for this purpose. From the study done by Yhdego et al. (2023), it was stated that the utilization of RNNs and LSTM models has unlocked new possibilities for analyzing and forecasting temporal data, particularly in the context of flight delay prediction. These models

are highly effective at forecasting propagation delays due to their ability to learn from time-series data.

According to Gülşen and Çetin (2024), one specialized type of RNN is the LSTM network. An LSTM network employs a unique memory cell structure combined with a set of gates—namely, the input gate, forget gate, and output gate—that meticulously regulate the flow of information. These mechanisms allow the network to selectively retain relevant data, discard unnecessary details, and control how stored information influences subsequent processing steps. This selective storage and retrieval capability enable LSTMs to manage sequential data with remarkable efficiency, making them highly suitable for solving complex problems involving time-dependent patterns.

According to Kilic and Sallan (2023), the majority of research studies address the challenge of predicting flight delays by treating it as a classification problem. In essence, these models are designed to ascertain whether a particular flight will experience a delay or not. The primary objective of these predictive models is to classify flights based on their likelihood of being delayed. The classification is typically determined by assessing whether the departure or arrival delay exceeds a specific threshold, which is commonly set at 15 minutes. Vandal et al., (2018) proposed a variational LSTM model to predict aggregate daily flight delays at U.S. airports. It leverages both continuous (weather, flight volumes, previous delays) and categorical features (airport, month, day) by embedding the categorical inputs. A three-layer LSTM architecture with Monte Carlo dropout was employed to perform approximate Bayesian inference, enabling uncertainty quantification. Experiments on four years of data from 123 airports yield a median absolute error of about 5.8 minutes per day.

RNNs that incorporate LSTM units have demonstrated remarkable effectiveness in handling tasks that necessitate the modeling and retention of long-term dependencies within sequential data. Among these applications, speech recognition stands out as one of the key areas where LSTM-equipped RNNs have achieved significant success. For instance, the work conducted by Bahdanau et al. (2014) highlights the potential of these advanced models in accurately capturing the nuances of spoken language and understanding temporal relationships in audio data. Similarly, the field of machine translation has greatly benefited from the deployment of LSTM-based RNNs.

III. Problem Description

This section presents an overview of the problem and data analysis conducted on the U.S airlines dataset obtained. Flight delays are a common issue caused by factors such

as adverse weather, air traffic control failures, navigation errors, and crew sicknesses. Accurate predictive models are essential for improving reliability and mitigating the impact of delays. According to the Federal Aviation Administration (FAA), flights landing over 15 minutes past their scheduled time are considered delayed. Delays often propagate to subsequent flights, exacerbating the issue. The growing demand for air travel has spurred significant research into flight delays. Predicting flight delays is crucial for enabling stakeholders to prepare and minimize the consequences.

This study analyses and uses a 12-month dataset of all the airports in the U.S and 28 operating U.S. airlines between January 2023 and December 2023. The dataset is obtained from the Bureau of Transportation Statistics, United States Department of Transportation. The datasets obtained were in two files. The first dataset file contained Weather data such as precipitation, wind speed, temperature, wind direction and snow levels, while the second dataset contained Flight data such as departure time, departure delay, arrival time, arrival delay, departure airport, arrival airport, airport congestion, aircraft age, flight duration and other relevant factors. The first dataset has a shape of 132,860 rows and 10 columns, while the second dataset has a shape of 6,743,404 rows and 24 columns. The flight delay data analysis is presented through bar plots and pie chart based on the selected features across the two dataset files used.

Figures 1 and 2 below are charts showing the diagrammatic representation of values correlating to the flight rate of the different airports represented in the datasets and the proportional contribution of Airlines with respect to their performing flight count to the whole datasets.

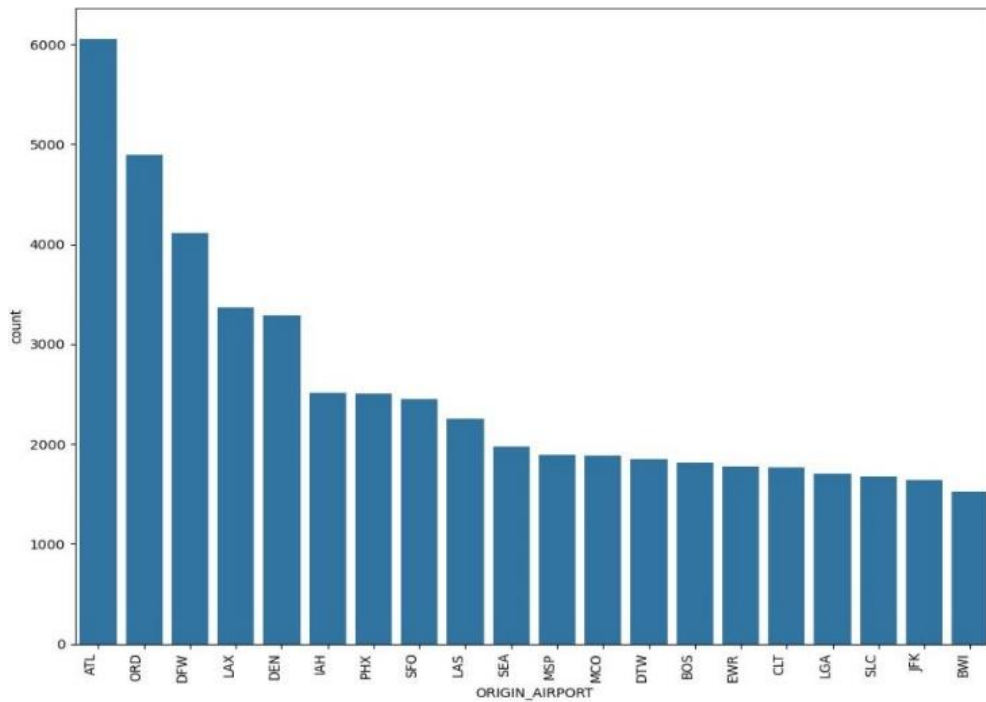


Fig. 1. Flight Rate of Different Airports

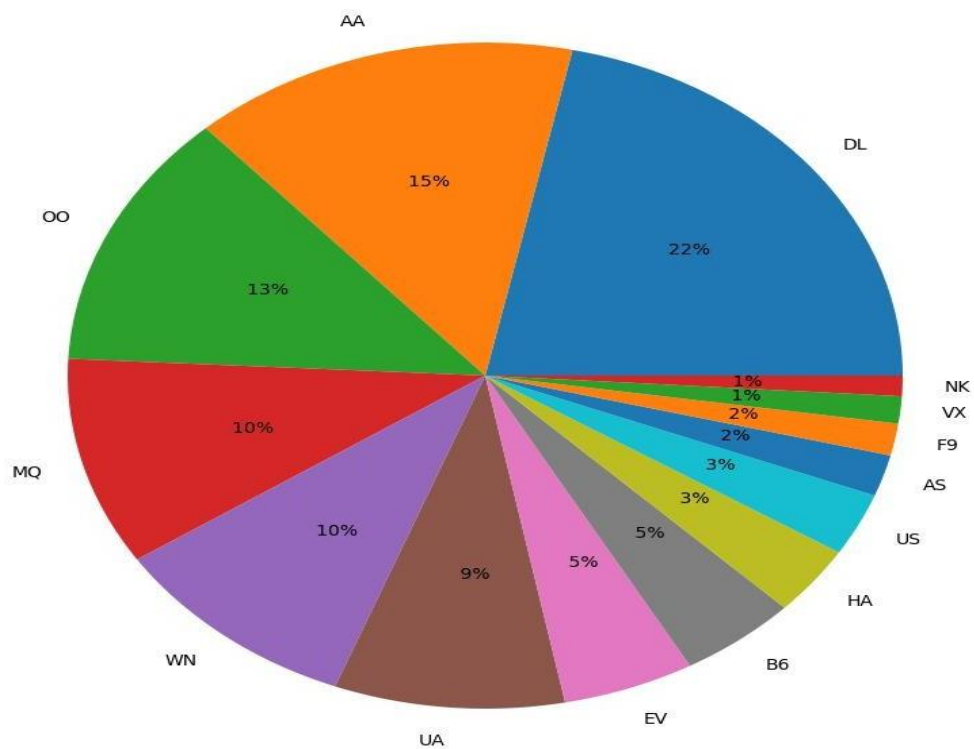


Fig. 2. Pie Chart View of Airlines with Their Performing Flight Count

IV. Methodology Applied

The primary goal of this project was to predict flight arrival delays by leveraging historical flight data and associated weather conditions. In our specific use case, a flight is considered delayed if its arrival delay in minutes is 15 minutes or more; otherwise, it is considered on time. This binary classification problem was approached using a deep learning model (an LSTM RNN) and hyperparameter optimization via a Genetic Algorithm (GA).

Integration Strategy:

The two datasets were merged.

After merging, we ensured that duplicate columns (which could occur during merging) were removed using a DataFrame expression that filters out duplicated column names.

Data Pre-processing

The pre-processing phase was critical to ensure that the data was in the proper format for deep learning:

- Target Variable Creation:

We transformed the continuous arrival delay into a binary target. Using a lambda function, any delay less than 15 minutes was set to 0 (on time), and any delay of 15 minutes or more was set to 1 (delayed).

- Handling Duplicates:

After merging, duplicate columns (including duplicate target columns) were removed to ensure consistency in feature space.

- Feature Selection:

We selected a set of features (including weather variables, flight characteristics, and airport identifiers) that were assumed to be predictive of flight delays.

- Categorical Encoding:

Categorical columns (such as airport codes, flight dates, delay types) were converted into numeric representations via ``LabelEncoder`` after filling missing values (with a placeholder like "Unknown"). This maintains consistency in training and later prediction.

- Filling Missing Values & Normalization:

- i. Numeric columns which had missing values were filled using their mean values.
- ii. The features were normalized using `MinMaxScaler`, which scales values between 0 and 1—this is important for effectively training a deep learning model like an LSTM.

- Splitting the Data:

We split the normalized data into training and testing sets (using an 80/20 ratio) to validate our model's performance.

- Reshaping for LSTM:

Since LSTM networks expect a 3D input (samples, timesteps, features), we reshaped our feature set by introducing an extra dimension (with a timestep of 1).

Model Design and Hyperparameter Optimization

- LSTM Model Construction:

The model architecture comprises:

- i. Two LSTM layers: the first LSTM layer returns sequences (useful for subsequent LSTM layers) and the second does not.
- ii. A dropout layer that helps reduce overfitting by randomly setting a fraction of the input units to 0 during training.
- iii. A dense layer with a sigmoid activation function that outputs a probability in a binary prediction task.

- Genetic Algorithm for Hyperparameter Tuning:

As the optimal capacity of the LSTM (the number of units) and the learning rate are essential for performance, PyGAD—a genetic algorithm framework—was used to perform hyperparameter optimization.

- i. The fitness function trains a model with a given set of hyperparameters for a few epochs and evaluates its test accuracy.
- ii. Genetic Algorithm settings (population size, number of generations) were defined, and the fitness function was updated to meet the API requirements of the latest PyGAD version.

Using the optimal hyperparameters found via GA, the final model is retrained on the entire training set.

The prediction phase involves converting the model's probability outputs into binary predictions (0 or 1), which are then evaluated using accuracy, precision, recall, and F1-score.

Model Evaluation

Once trained, the final model was evaluated using standard metrics:

- i. Accuracy
- ii. Precision
- iii. Recall
- iv. F1-score

These metrics provide insights into both the overall performance and the model's balance between false positives and false negatives.

Deployment-Readiness: Saving the Model and Preprocessors

To move the model into production for everyday predictions:

- i. The final trained LSTM model was saved using Keras's native format.
- ii. Essential preprocessing objects (the MinMaxScaler and dictionary of LabelEncoders) were persisted using Python's `pickle` module. This ensures that any new data can be preprocessed consistently with the training data.

A separate prediction script is created. This script loads the saved model and preprocessing objects.

New incoming data is preprocessed exactly as the training data (using the saved scaler and label encoders).

The preprocessed data is reshaped and fed to the model to generate predictions.

For correctness, the model's binary predictions 0 or 1 are mapped to labels "Delayed" and "On-Time" respectively before being presented.

V. Implementation Results

Figure 3 below shows the models progress during training with the selected datasets.

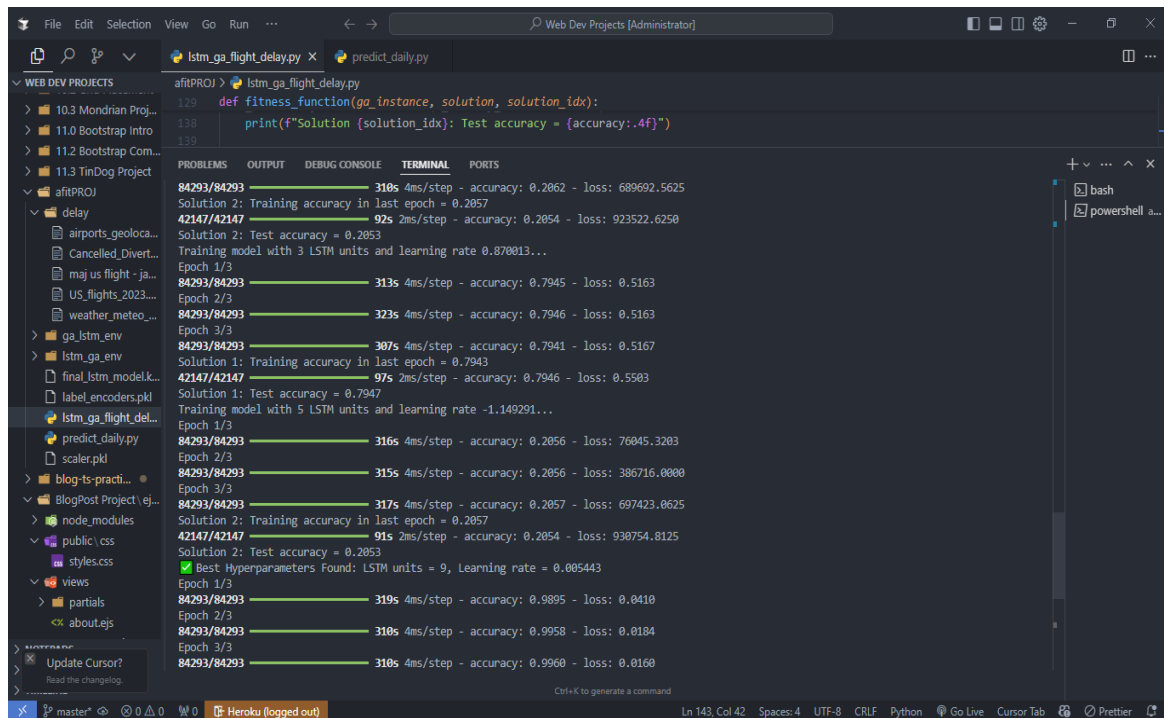


Fig. 3. A Snapshot of the Terminal-Logs of the model during Training

Figure 4 below shows the results of the metrics evaluation after the models training is completed.

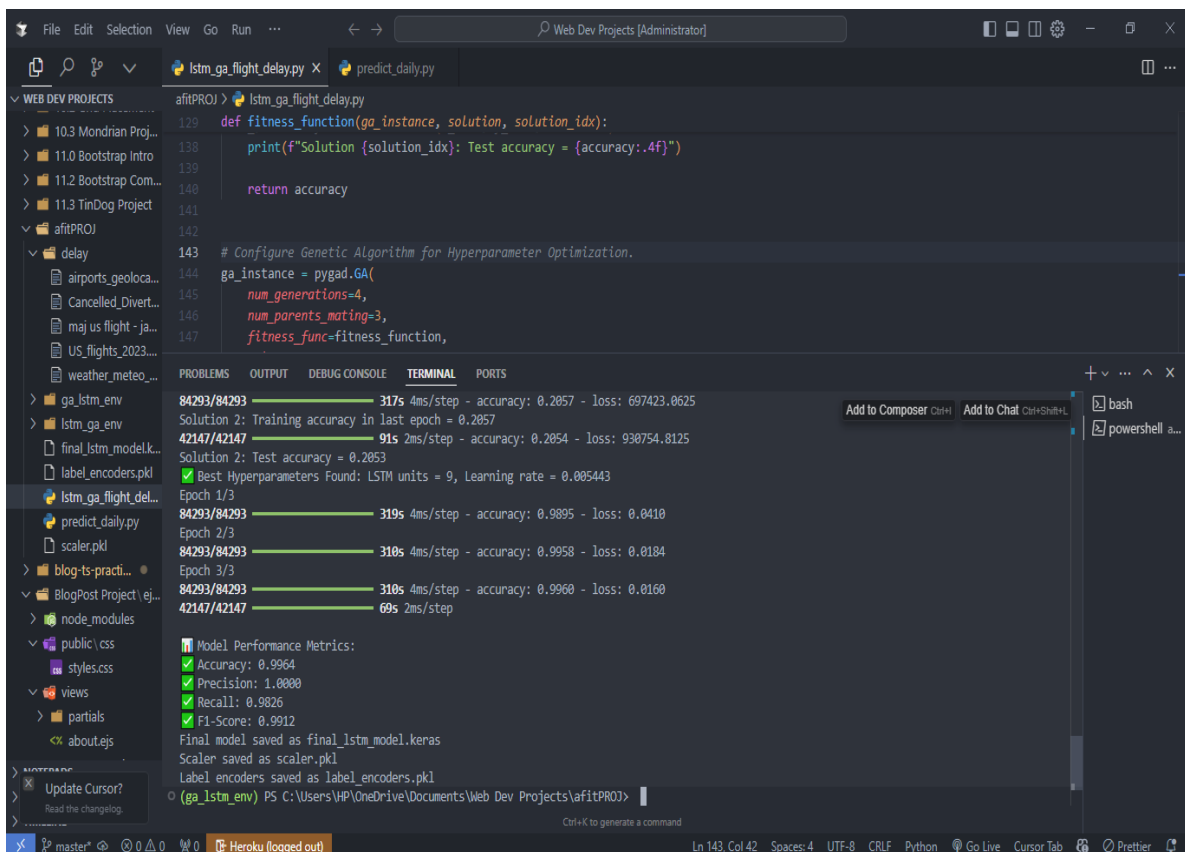


Fig. 4. A Snapshot of the Result of the Metrics Evaluation

The trained model was able to give a result of 99% of accuracy, 100% of precision, 98% in the recall and 99% in the f1-score. The model performed extremely well and its result were based on the dataset that was used which for sure had a large presence of outliers.

VI. Discussion, Conclusion and Future Study

Discussion

The study of flight delays in the context of U.S. airlines shows critical insights into the challenges faced by the aviation industry to manage and mitigate delays. The approach proposed for flight delays in this research, which combines LSTM RNN with Genetic Algorithm, demonstrates significant improvements to predict flight delays compared to traditional methods. The effectiveness of this work is expressed by handling complex and large-scale datasets. The higher accuracy, precision, recall and F1-score, in the case of departure delay, arrival delay, and total delay classification, are considerably impressive.

The analysis of flight delay patterns across different airlines, states, and times intervals provides valuable insights for operational decision-making. Airlines can also contact passengers and update them continuously regarding their flights. In this case, if the passengers are in the know of their flight status, then they can manage their next schedule appropriately. Overall, this approach can reduce the anxiety level of passengers as well as lower their expenses, since sudden changes in schedule can add to their trip cost. Similarly, the analysis applies for the flight arrival delay problem.

Conclusion and Future Study

The project successfully gave a predictive model in terms of flight delays using machine learning. It made use of the LSTM-RNN algorithm, it proved to be effective, with the model aim of the project which is been achieved. It also went further in giving a high evaluation metrics. The project also provided the importance of factors such as weather and departure delays in the prediction of flight delays. The developed manual testing function showed how the model can be put in practice for further enhancement.

At the end of the implementation of project, it was understood that more works can be done which include; the integration of real-time weather data and also current flight information to give the model its accuracy and applicability of real life application. Also experimenting with other machine learning algorithms like the XGBoost, or probably the expansion of feature engineering and the integration of real world flight data APIs. Performing a more detailed exploratory data analysis in order to uncover deeper insights to the causes of delays, putting in more contextual and external factors as well.

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